

Submit your Jupyter notebook with the answers to gradescope by 1:50pm.

1. Make a fast callable object of $\frac{\cos(x) + x^2 + y}{x + y^2}$ and draw the expression tree.
2. Consider the function $f(n) = \frac{1}{n} \sum_{k=1}^{n-1} \sin\left(\pi \frac{k}{n}\right)$.
 - (a) Use the above definition of f to make a function **F** which for some positive integer n returns the floating-point approximation of $f(n)$ in double precision.
 - (b) Time the execution of **F** for $n = 10,000$. Explain why **F** is inefficient.
 - (c) Define the function **V**, vectorizing **F** with numpy. Verify its correctness. Time the execution of **V** for $n = 10,000$ and compare with timings for **F**.